**COMP2111 Database Design**

**Class Practice 8 Handling Referential Integrity**

Part I: Build the following table structure and integrity constraints in MS Access.

Let's simply use short text, size 3 as the domain for the attributes.

Staff(staffNo, sName)

Client(clientNo, staffNo)

FK staffNo references Staff(staffNo) – On Update Cascade; On Delete No Action

Property(propertyNo, staffNo)

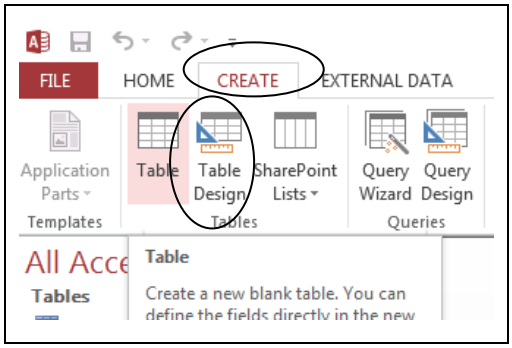
FK staffNo references Staff(staffNo) – On Update Cascade; On Delete No Action

Viewing(clientNo, propertyNo, comment)

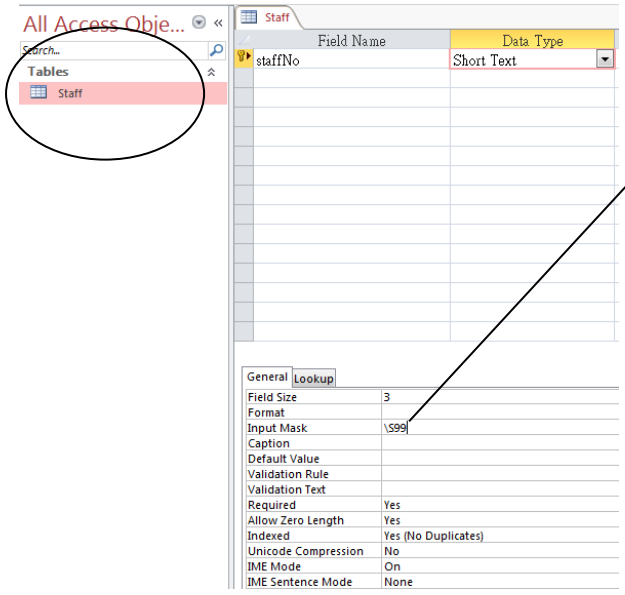
FK clientNo references Client (clientNo) – On Update Cascade; On Delete No Action

FK propertyNo references Property (propertyNo) – On Update Cascade; On Delete Cascade

**Create → Table Design**




After table is created, you can always **right click on the table name to select "Design View"** to look at the table structure.



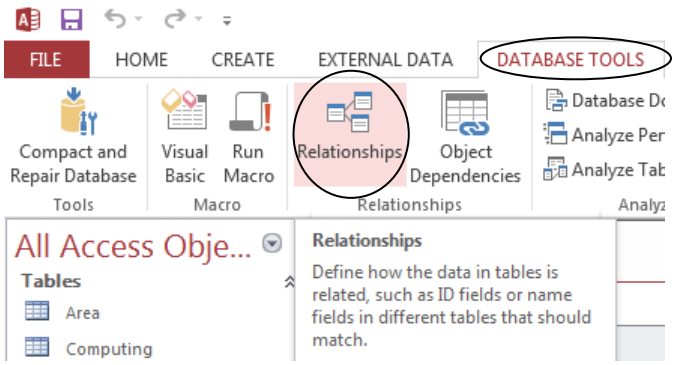

This input masks means must start with "S" with maximum of 2 digits following it. For details of input mask, refer to http://office.microsoft.com/en-us/access-help/control-data-entry-formats-with-input-masks-HA010096

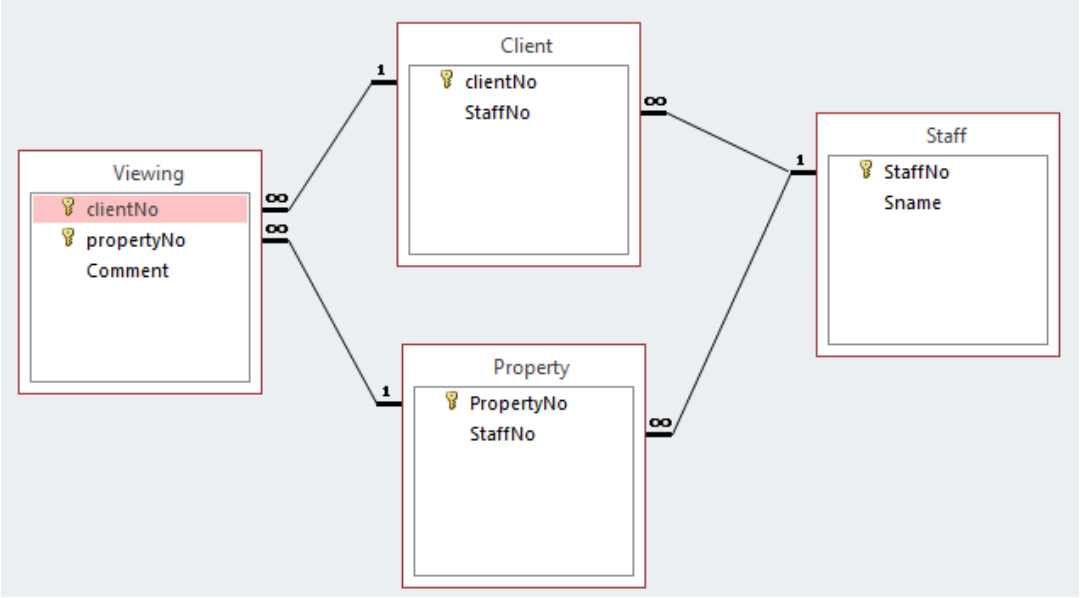
Note that if you apply this, you have to apply it to all other foreign keys of staffNo.

After building the table structure, let's work on the relationship between the tables.

**Go to Database Tools → Relationships to create the relationship and the integrity constraints.**

**Make sure that all the tables are closed before you work on the relationship.**



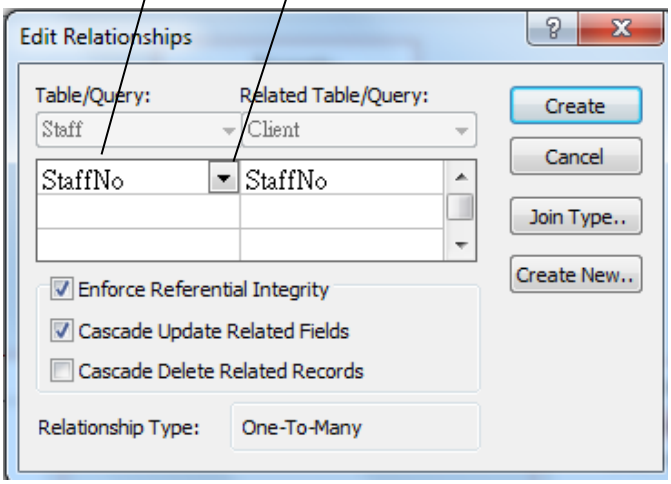




Make sure data in these two columns are correct

Enforce Referential Integrity:
- If Checked: No action;
- If Unchecked: No check.

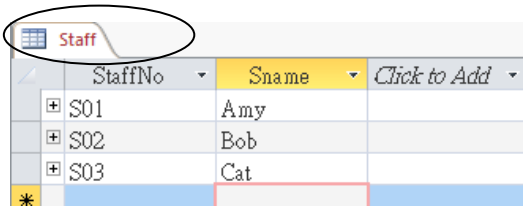
Cascade on Update:
- If Checked: Cascade;
- If Unchecked: No action.

Cascade on Delete:
- If Checked: Cascade;
- If Unchecked: No action.

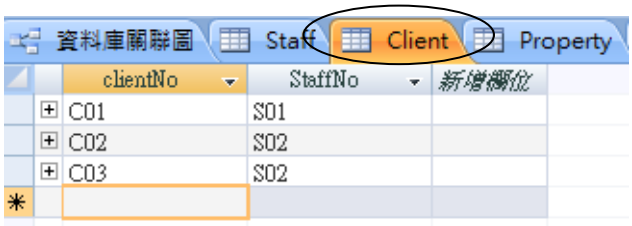


**Part II: Input the data into the tables accordingly.**

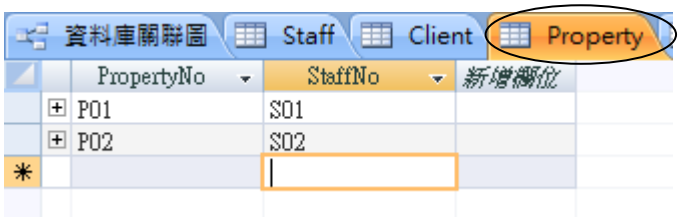
1. Inserting records into Staff (Parent table) is always successful without violating referential integrity rule.



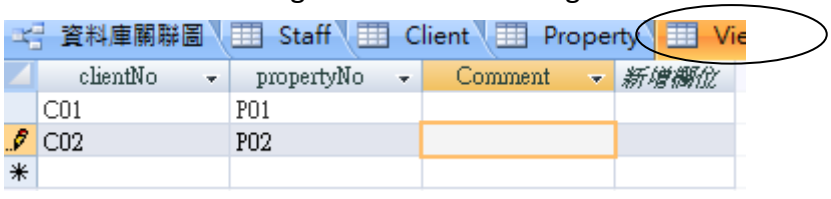

2. Insert record C04 into Client (Child) with StaffNo S99. Can it be done and what is the reason?




3. Insert record P03 with staffNo being null. Can it be done and what is the reason?




4. Insert the following records into Viewing table.




Note that because of the setting of referential integrity (RI), you can't input a record that violates referential integrity.

**Part III: Answer the following questions.**

1. This question is based on **Staff & Client's Parent-child relationship**:

a) Delete record S01 from Staff (Parent), what happens and why? But why can you delete S03?

b) Delete record C03 from Client (Child), what happens and why?

c) Update S01 of Staff (Parent) to S05, what happens to the corresponding record in the Client (Child)? Why?

d) Update the referential integrity between Staff and Client by deleting the "On Update Cascade". Then repeat question (c) above. What happens and why?

2. This question is based on Viewing entity:

a) Delete record P02 from Property (Parent of Viewing), what happens to the corresponding child record in Viewing? Why?